Identification strategy

Our identification strategy relies on focusing on families of up to three children and focusing on parents with twin daughter. We focus on families of up to three children as larger families might be distinct for various other reasons that could confound the analysis. We focus on twin daughters for identification strategy as because having twin daughters is a quasi-experimental shock of two additional daughters, while an additional single daughter might come from household family planning and, thus, plausibly less of a 'shock'.

We considered additional control to include like number of sisters of the respondent, marital status, and whether the family lives together in one household or not. Given the limited sample leftover through the identification strategy of focusing on twin daughter we elect to not control for these.

Hypothesis 1

Variables:

- For attitudes of gender equality, will use survey question: *Best if man and woman work the same amount so they can share the responsibility* (variable name: plh0308_v2). Values less than 0 are either non-answers or not part of the survey and so will be excluded
- For twin daughters, will use variables that identify gender of kids (kidsex01-kidsex03) and construct a dummy for having twin daughters using the birthyear of the respondent's children (kidgeb01- kidgeb03)
- To control for income, we will use the variable on gross labour income (pglabgro) and transform by taking logs
- To control for education, we will use the variable on amount of education in years (pgbilzeit) truncated for values above 0 as negative values are for missing cases
- To control for age, we will use the age variable (bioage) restricting to observations greater than 0 as negative values are for missing cases

Inclusion/exclusion criteria:

- Focus on private households as it is not clear what the other categories are referring to: keep if inlist(pop,1,2)
- Focus on families with three children or less. This is because larger families are potentially distinct in various ways that we do not want to confound the analysis

Exact analysis:

As the survey item we use as outcome variable is only included in one year, this is a simple OLS regression.

$$Gender_Equality_i = \beta_0 + \beta_1 TwinDaughters_i + \beta_2 Log_Income_i + \beta_3 Education_i + \beta_4 Age_i + \epsilon_i$$

Using clustered errors at region of birth (birthregion)

Interpretation of results:

- Check the direction and magnitude of β_1. A positive estimate is in line with the hypothesis
- Check for statistical significance from the t-test p-value of the regression output

Hypothesis 2

Variables:

- For political preferences, will use survey question: *Best if man and woman work the same amount so they can share the responsibility* (variable name: plh0004). Values less than 0 are either non-answers or not part of the survey and so will be excluded
- For twin daughters, will use variables that identify gender of kids (kidsex01-kidsex03) and construct a dummy for having twin daughters using the birthyear of the respondent's children (kidgeb01- kidgeb03)
- To control for income, we will use the variable on gross labour income (pglabgro) and transform by taking logs
- To control for education, we will use the variable on amount of education in years (pgbilzeit) truncated for values above 0 as negative values are for missing cases
- To control for age, we will use the age variable (bioage) restricting to observations greater than 0 as negative values are for missing cases

Inclusion/exclusion criteria:

- Focus on private households as it is not clear what the other categories are referring to: keep if inlist(pop,1,2)
- Focus on families with three children or less. This is because larger families are potentially distinct in various ways that we do not want to confound the analysis: keep if kidgeb04<0

Exact analysis:

Panel regression with time (α_t) and birth region (η_r) fixed effects:

$$Political\ Preference_{it} = \beta_0 + \beta_1 TwinDaughters_{it} + \beta_2 Log_Income_{it} + \beta_3 Education_{it} + \beta_4 Age_{it} + \alpha_t + \eta_r + \epsilon_{it}$$

Using clustered errors at region of birth (birthregion).

Interpretation of results:

- Check the direction and magnitude of β_1. A negative estimate is in line with the hypothesis
- Check for statistical significance from the t-test p-value of the regression output

Hypothesis 3

Variables:

- For gender equality in the household, measured by behaviours or outcomes, will use survey question: *Hausarbeit Std./Werktag [1992-2020]* (variable name: pli0043_v3). Values less than 0 are either non-answers or not part of the survey and so will be excluded
- For twin daughters, will use variables that identify gender of kids (kidsex01-kidsex03) and construct a dummy for having twin daughters using the birthyear of the respondent's children (kidgeb01- kidgeb03)
- To control for income, we will use the variable on gross labour income (pglabgro) and transform by taking logs
- To control for education, we will use the variable on amount of education in years (pgbilzeit) truncated for values above 0 as negative values are for missing cases
- To control for age, we will use the age variable (bioage) restricting to observations greater than 0 as negative values are for missing cases

Inclusion/exclusion criteria:

- Focus on private households as it is not clear what the other categories are referring to: keep if inlist(pop,1,2)
- Focus on families with three children or less. This is because larger families are potentially distinct in various ways that we do not want to confound the analysis: keep if kidgeb04<0
- Focus on male respondents, this way larger values of housework is indicative of more gender equal behaviours within the household

Exact analysis:

Panel regression with time (α_t) and birth region (η_r) fixed effects:

$$Household\ Equality_{it} = \beta_0 + \beta_1 TwinDaughters_{it} + \beta_2 Log_Income_{it} + \beta_3 Education_{it} + \beta_4 Age_{it} + \alpha_t + \eta_r + \epsilon_{it}$$

Using clustered errors at region of birth (birthregion)

Interpretation of results:

- Check the direction and magnitude of β_1. A positive estimate is in line with the hypothesis
- Check for statistical significance from the t-test p-value of the regression output

Hypothesis 4

Variables:

- For attitudes of gender equality, will use survey question: *Best if man and woman work the same amount so they can share the responsibility* (variable name: plh0308_v2). Values less than 0 are either non-answers or not part of the survey and so will be excluded
- For twin daughters, will use variables that identify gender of kids (kidsex01-kidsex03) and construct a dummy for having twin daughters using the birthyear of the respondent's children (kidgeb01- kidgeb03)
- Use variable for gender (sex) to create dummy for male respondent
- To control for income, we will use the variable on gross labour income (pglabgro) and transform by taking logs
- To control for education, we will use the variable on amount of education in years (pgbilzeit) truncated for values above 0 as negative values are for missing cases
- To control for age, we will use the age variable (bioage) restricting to observations greater than 0 as negative values are for missing cases

Inclusion/exclusion criteria:

- Focus on private households as it is not clear what the other categories are referring to: keep if inlist(pop,1,2)
- Focus on families with three children or less. This is because larger families are potentially distinct in various ways that we do not want to confound the analysis: keep if kidgeb04<0

Exact analysis:

As the survey item we use as outcome variable is only included in one year, this is a simple OLS regression.

$$Gender_Equality_i = \beta_0 + \beta_1 TwinDaughters \times Male_i + \beta_2 TwinDaughters_i + \beta_3 Male_i + \beta_4 Log_Income_i + \beta_5 Education_i + \beta_6 Age_i + \epsilon_i$$

Using clustered errors at region of birth (birthregion)

Interpretation of results:

- Check the direction and magnitude of β_1. A positive estimate is in line with the hypothesis
- Check for statistical significance from the t-test p-value of the regression output